Board Game Ratings Analysis

DSCI-510 Final Presentation

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Introduction

- This project analyzes factors that influence board game ratings using machine learning techniques.
- The goal is to understand how features such as playtime, player count, and complexity relate to user ratings.

Data sources

- **BoardGameGeek Ranking Dataset**
- https://boardgamegeek.com/data_dumps/bg_ranks
- An official dataset containing board game rankings, average ratings, and user statistics.
  This dataset is used to identify the top 500 board games for analysis.

Daniel Karp @dakarp · 2023年10月21日 (edited)

I've added process to generate a daily CSV you can use to download all of the games in the database, with their names, ids, ranks, ratings, and number of raters. You can find it here.
https://boardgamegeek.com/data_dumps/bg_ranks

As a reminder, scraping our site, aside from those endpoints we explicitly provide (the XML API), is against our terms of service.
For licensing purposes, this data is considered part of our XML API. See our terms of use for the XML API.

24 3 Reply Quote ···

Data sources

- **BoardGameGeek XML API**
- https://boardgamegeek.com
- Used to retrieve detailed game information, including mechanics, categories, player count, playtime, and complexity.
  This serves as the primary source of features for the machine learning model.

	id	name	yearpublished	rank	bayesaverage	average	usersrated	is_expansion	abstracts_
0	224517	Brass: Birmingham	2018	1	8.39390	8.56536	57997	0	
1	342942	Ark Nova	2021	2	8.35499	8.54101	60423	0	
2	161936	Pandemic Legacy: Season 1	2015	3	8.34777	8.50412	57266	0	
3	174430	Gloomhaven	2017	4	8.29966	8.54115	67001	0	
4	397598	Dune: Imperium – Uprising	2023	5	8.23216	8.70180	17936	0	

Data sources

- **Kaggle Dataset**
- https://www.kaggle.com/datasets/sujaykapadnis/board-games
- A dataset derived from the BoardGameGeek API, containing board game attributes and rating-related information. Used for initial exploration to understand the structure of board game data.

Board Games Dataset

Board Games Dataset from boardgamegeek.com

Data Card | Code (5) | Discussion (0) | Suggestions (0)

About Dataset

This data comes from the Board Game Geek database. The site's database has more than 90,000 games, with crowd-sourced ra
also an R package with the fulldataset (`bggAnalysis`) but it hasn't been updated in ~2 years.

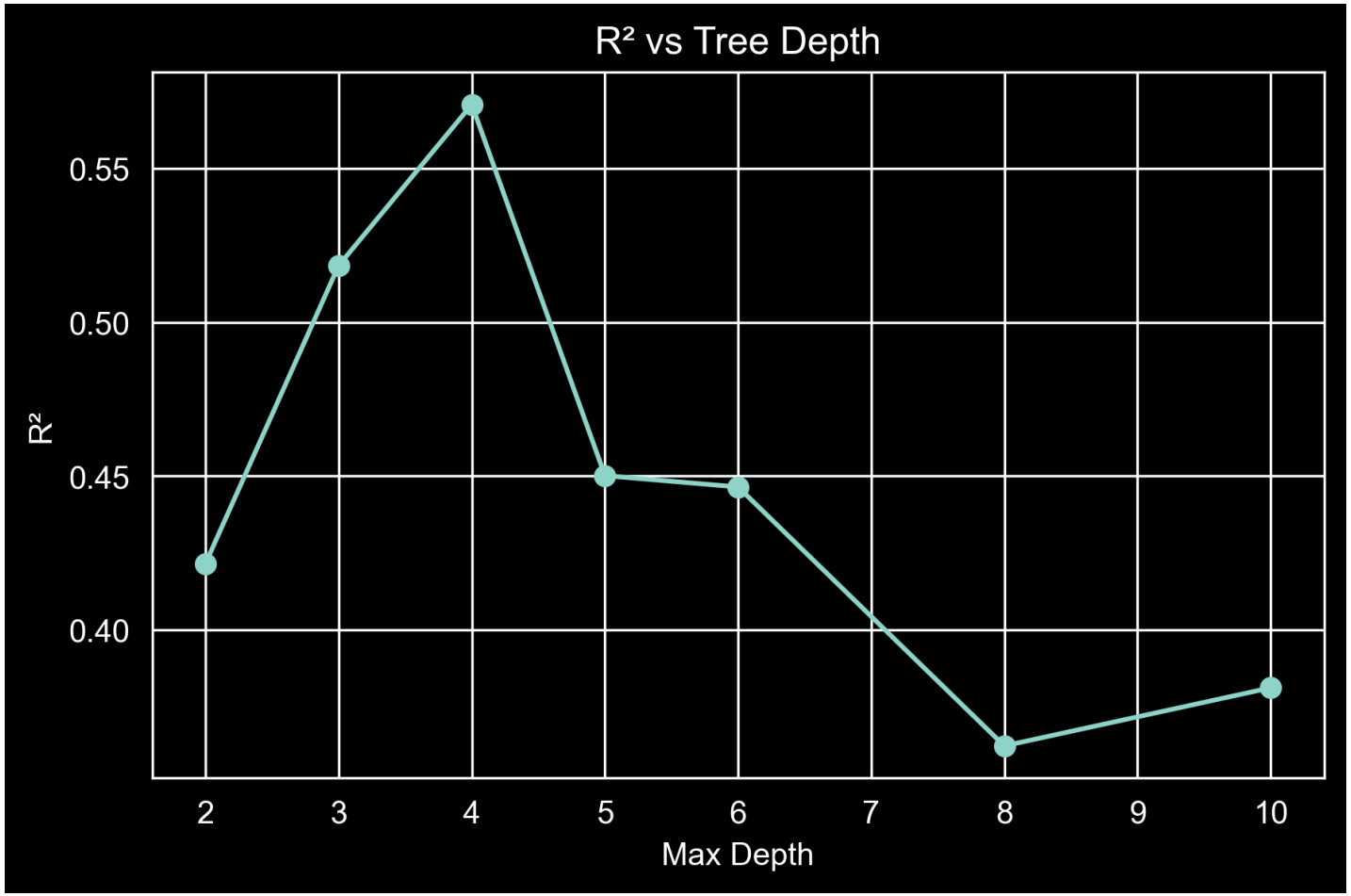
Data Dictionary

variable	class	description
game_id	character	Unique game identifier
description	character	A paragraph of text describing the game
image	character	URL image of the game
max_players	integer	Maximum recommended players
max_playtime	integer	Maximum recommended playtime (min)

Results

Model Performance vs Tree Depth
RMSE
0.26
0.25
0.24
0.23
0.22
2
3
4
5
6
7
8
9
10
Max Depth

Results



Results

numweights	0.372351
yearpublished	0.338548
Push Your Luck	0.066093
minage	0.060921
Variable Set-up	0.047283
playingtime	0.045378
Medical	0.019033
Action / Dexterity	0.018908
maxplaytime	0.017231
Tags	0.014253

Results

```
Push Your Luck                          0.042727
Variable Set-up                         0.031913
Increase Value of Unchosen Resources    0.025997
Tile Placement                          0.010312
Delayed Purchase                        0.009515
Tags                                    0.009213
Lose a Turn                             0.009061
Trick-taking                            0.008913
Storytelling                            0.008289
Stat Check Resolution                   0.007564
dtype: float64
```

Challenges

- BGG policy restrictions on web scraping
- API access approval delay (~2 weeks)
- API rate limiting (batch requests required)

Thank you!